Solution Problem 11
Mechanics of two or more particles

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Oct 14, 2017

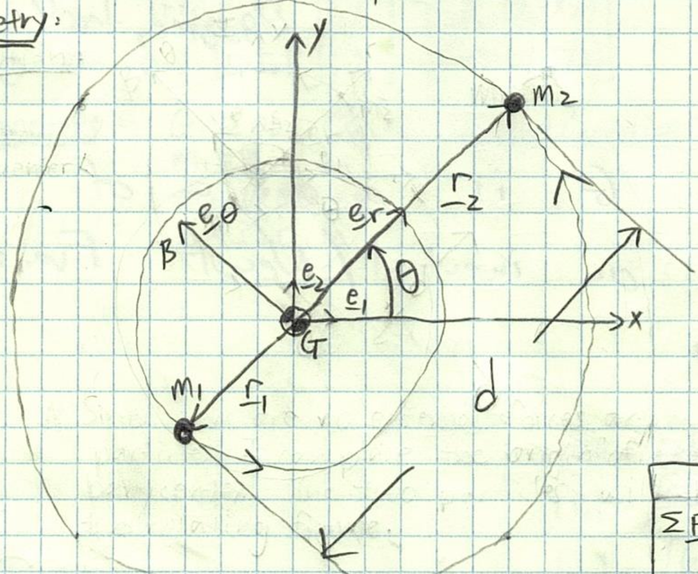
(a) **Problem**: Two particles of unequal masses are orbiting each other. What is the period of circular motion?

Assumptions
- inverse square Newtonian gravity: $F = \frac{Gm_1m_2}{r^2}$
- circular motion
- no external forces

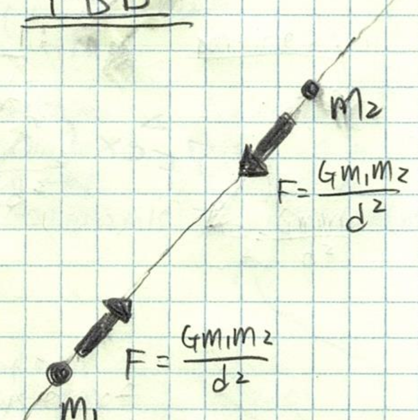
Parameters:
- G = gravitational constant
- m_1 = mass of particle 1
- m_2 = mass of particle 2
- d = distance between the two particles

Geometry:

G = COM



FBD



Calculate $|\underline{r}_1|$ and $|\underline{r}_2|$

$\Sigma \underline{F}^{(ext)} = \underline{0}$ implies G is origin of inertial frame

$|\underline{r}_1| + |\underline{r}_2| = d$

$m_1\underline{r}_1 + m_2\underline{r}_2 = m_{tot}\underline{r}_G = 0$ (COM definition)

$\underline{r}_1 = -\frac{m_2}{m_1}\underline{r}_2$

$|\underline{r}_1| = \frac{m_2}{m_1}|\underline{r}_2| \Rightarrow \frac{m_2}{m_1}|\underline{r}_2| + |\underline{r}_2| = d.$

$|\underline{r}_2| = \frac{m_1}{m_1+m_2}d, \; |\underline{r}_1| = \frac{m_2}{m_1+m_2}d$

circular motion $\Rightarrow \ddot{r}_1 = \dot{r}_1 = 0$

$\underline{\text{LMB}_1}$: $\Sigma \underline{F}_1 = m_1\underline{a}_1$

$$\frac{G\cancel{m_1}m_2}{d^2}\underline{e}_r = \cancel{m_1}[(\cancel{\ddot{r}_1}^{\,0} - r_1\dot{\theta}^2)\underline{e}_r + (r_1\ddot{\theta} + \underbrace{\cancel{2\dot{r}_1\dot{\theta}}}_{0})\underline{e}_\theta]$$

dot with $\underline{e}_r \Rightarrow \frac{Gm_2}{d^2} = r_1\dot{\theta}^2 = \frac{m_2}{m_1+m_2}d\,\dot{\theta}^2$

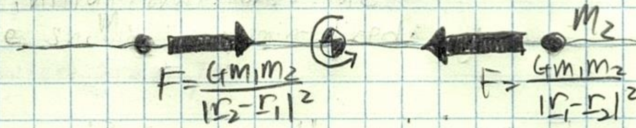
$$\dot{\theta} = \sqrt{\frac{G(m_1+m_2)}{d^3}}$$

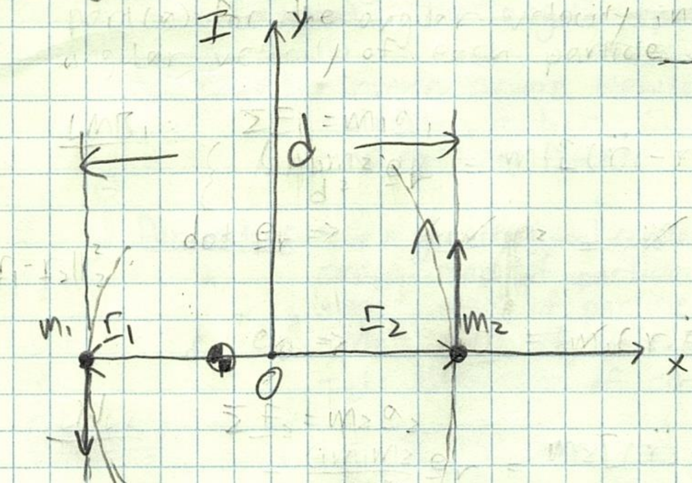
dot with $\underline{e}_\theta \Rightarrow 0 = r_1\ddot{\theta} \Rightarrow \ddot{\theta} = 0 \Rightarrow$ angular velocity is constant.

$$\text{period} = \frac{2\pi}{\dot{\theta}} = \boxed{2\pi\sqrt{\frac{d^3}{G(m_1+m_2)}}}$$

(b) Integrate LMB equations in cartesian using ode45.

FBD





LMB_1: $\Sigma \underline{F}_1 = m_1\underline{a}_1$

$$\frac{Gm_1m_2}{|\underline{r}_2-\underline{r}_1|^3}(\underline{r}_2-\underline{r}_1) = m_1(\ddot{x}\underline{e}_1 + \ddot{y}\underline{e}_2)$$

dot $\underline{e}_1 \Rightarrow$ $$\frac{Gm_1m_2}{|\underline{r}_2-\underline{r}_1|^3}(x_2-x_1) = m_1\ddot{x}_1$$

dot $\underline{e}_2 \Rightarrow$ $$\frac{Gm_1m_2}{|\underline{r}_2-\underline{r}_1|^3}(y_2-y_1) = m_1\ddot{y}_1$$

Same process to obtain EOM for mass 2.

$$\frac{Gm_1m_2}{|\underline{r}_1-\underline{r}_2|^3}(x_1-x_2) = m_2\ddot{x}_2$$

$$\frac{Gm_1m_2}{|\underline{r}_1-\underline{r}_2|^3}(y_1-y_2) = m_2\ddot{x}_2$$

$$\begin{bmatrix} |\underline{r}_2-\underline{r}_1|^3 & 0 & 0 & 0 \\ 0 & |\underline{r}_2-\underline{r}_1|^3 & 0 & 0 \\ 0 & 0 & |\underline{r}_1-\underline{r}_2|^3 & 0 \\ 0 & 0 & 0 & |\underline{r}_1-\underline{r}_2|^3 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \\ \ddot{x}_2 \\ \ddot{y}_2 \end{bmatrix} = \begin{bmatrix} Gm_2(x_2-x_1) \\ Gm_2(y_2-y_1) \\ Gm_1(x_1-x_2) \\ Gm_1(y_1-y_2) \end{bmatrix}$$

Parameters:
$G = 2000\ m^3/kgs^2$
$m_1 = 800$ kg
$m_2 = 400$ kg
$d = 300$ m

Initial Conditions:

$x_{10} = -150$ m	$x_{20} = 150$ m
$y_{10} = 0$ m	$y_{20} = 0$ m
$\dot{x}_{10} = 0$ m/s	$\dot{x}_{20} = 0$ m
$\dot{y}_{10} = -29.8142$ m/s	$\dot{y}_{20} = 59.6285$ m/s

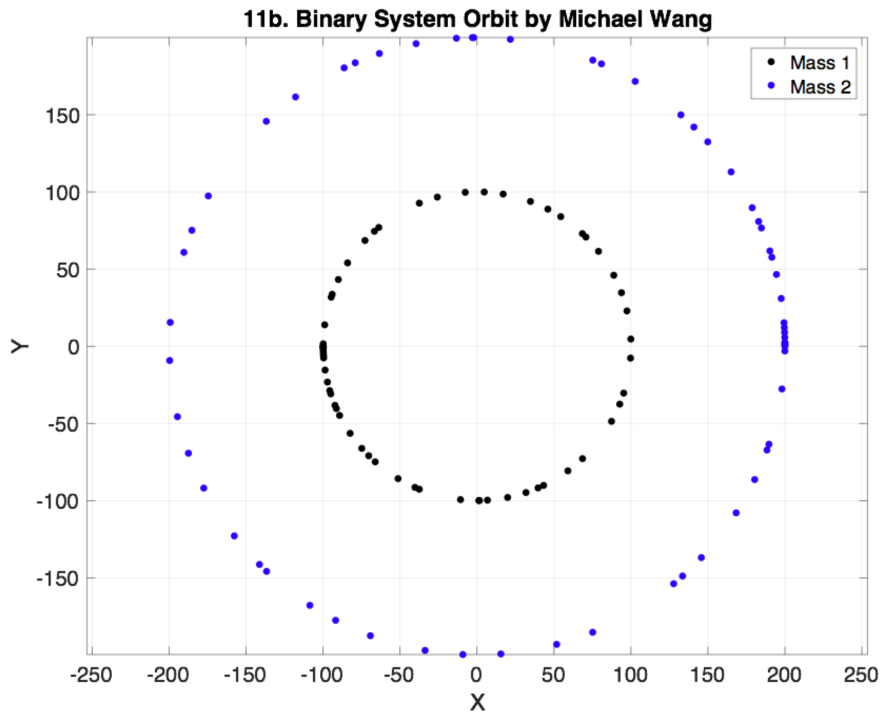
$$|\underline{v}| = \dot{\theta}\, \overset{\text{moment arm about COM}}{r}$$

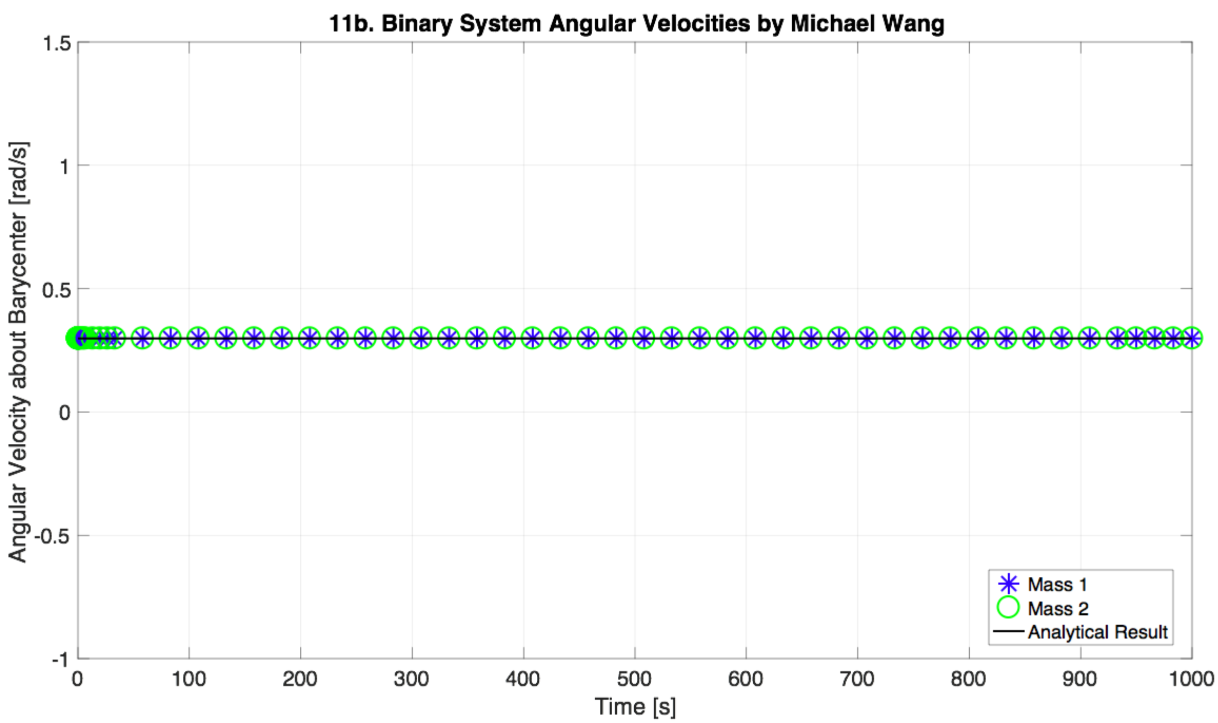
$$|v_1| = \sqrt{\frac{G(m_1+m_2)}{d^3}}\,\frac{m_2}{m_1+m_2}d = 29.8142\ m/s$$

$$|v_2| = \sqrt{\frac{G(m_1+m_2)}{d^3}}\,\frac{m_1}{m_1+m_2}d = 59.6285\ m/s$$

As shown in MATLAB plot, the angular velocities of the two bodies about COM correspond to value in part (a).

Analytical	Simulation
0.298 rad/s	0.298 rad/s


11b. Binary System Orbit by Michael Wang
Mass 1
Mass 2
150
100
50
0
-50
-100
-150
Y
-250
-200
-150
-100
-50
0
50
100
150
200
250
X


11b. Binary System Angular Velocities by Michael Wang
1.5
1
0.5
0
-0.5
-1
Angular Velocity about Barycenter [rad/s]
Mass 1
Mass 2
Analytical Result
0
100
200
300
400
500
600
700
800
900
1000
Time [s]

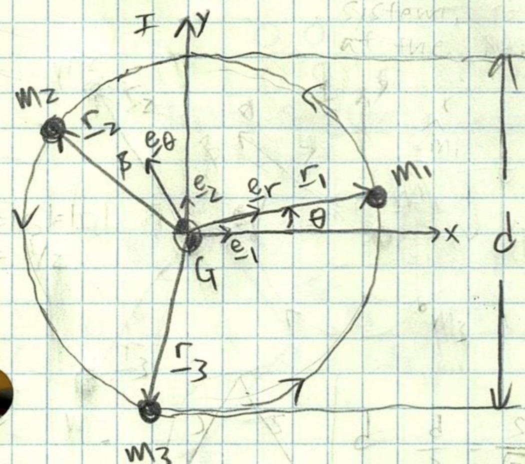
(c) Problem: Three equal mass particles orbiting about barycenter. What is the angular speed of circular motion?

Assumptions: - inverse square Newton gravity
- circular motion
- no external forces

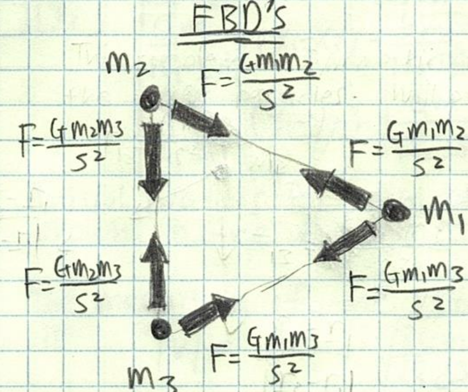
Parameters: - G = gravitational constant = 10
- m_1 = mass of particle 1 = 1
- m_2 = mass of particle 2 = 1
- m_3 = mass of particle 3 = 1
- d = diameter of circular orbit = 3

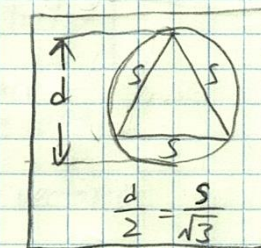
Geometry:

$\sum \underline{F}^{(ext)} = 0 \Rightarrow$ origin of inertial frame at center of mass



FBD'S





LMB$_1$: $\sum \underline{F}_1 = m_1 \underline{a}_1$

$$\underline{F}_{12} + \underline{F}_{13} = \frac{Gm_1m_2}{s^2}\left(\frac{\underline{r}_2-\underline{r}_1}{s}\right) + \frac{Gm_1m_3}{s^2}\left(\frac{\underline{r}_3-\underline{r}_1}{s}\right) = m_1\underline{a}_1$$

$\boxed{\text{Circular motion } \ddot{r}_1 = \dot{r}_1 = 0}$

let $m_1=m_2=m_3=m$

$$\frac{Gm^{\cancel{2}}}{s^3}(\underline{r}_2-\underline{r}_1) + \frac{Gm^{\cancel{2}}}{s^3}(\underline{r}_3-\underline{r}_1) = \cancel{m}\left[(\overset{0}{\cancel{\ddot{r}_1}} - r_1\dot{\theta}_1^2)\underline{e}_r + (r_1\ddot{\theta}_1 + \overset{0}{\cancel{2\dot{r}_1\dot{\theta}_1}})\underline{e}_\theta\right]$$

dot $\underline{e}_r \Rightarrow$ $$\frac{Gm}{s^3}\left(\frac{d}{2}\cos(120^\circ) - \frac{d}{2}\right) + \frac{Gm}{s^3}\left(\frac{d}{2}\cos(120^\circ) - \frac{d}{2}\right) = -\frac{d}{2}\dot{\theta}_1^2$$

$$\frac{Gm}{(\sqrt{3}d/2)^3}(3) = \dot{\theta}_1^2$$

dot $\underline{e}_\theta \Rightarrow$ $$\frac{Gm}{s^3}\left(\frac{d}{2}\cos 30^\circ - 0\right) + \frac{Gm}{s^3}\left(\frac{d}{2}\cos(150^\circ) - 0\right) = r_1\ddot{\theta}_1$$

$0 = r_1\ddot{\theta}_1 \Rightarrow \ddot{\theta}_1 = 0 \Rightarrow$ constant angular velocity

$$\dot{\theta}_1 = \dot{\theta}_2 = \dot{\theta}_3 = \boxed{\sqrt{\frac{8Gm}{\sqrt{3}d^3}}}$$

(d) Use ode45 to integrate dynamics for part(c). Use result from part c for the initial condition of the angular velocity.

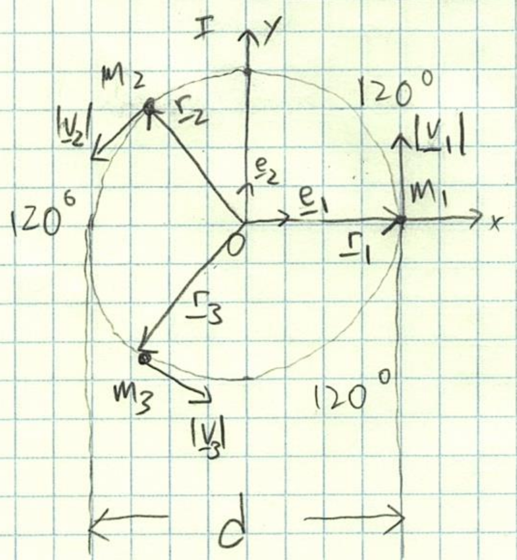
Using LMB from before, the equations of motion in Cartesian:

$$\boxed{\begin{aligned}\ddot{x}_1 &= \frac{Gm}{|\underline{r}_2-\underline{r}_1|^3}(x_2-x_1) + \frac{Gm}{|\underline{r}_3-\underline{r}_1|^3}(x_3-x_1)\\ \ddot{y}_1 &= \frac{Gm}{|\underline{r}_2-\underline{r}_1|^3}(y_2-y_1) + \frac{Gm}{|\underline{r}_3-\underline{r}_1|^3}(y_3-y_1)\end{aligned}}$$

repeat for other 2 masses

$$\begin{bmatrix}1&0&0&0&0&0\\0&1&0&0&0&0\\0&0&1&0&0&0\\0&0&0&1&0&0\\0&0&0&0&1&0\\0&0&0&0&0&1\end{bmatrix}\begin{bmatrix}\ddot{x}_1\\\ddot{y}_1\\\ddot{x}_2\\\ddot{y}_2\\\ddot{x}_3\\\ddot{y}_3\end{bmatrix}=\begin{bmatrix}\frac{Gm}{|\underline{r}_2-\underline{r}_1|^3}(x_2-x_1)+\frac{Gm}{|\underline{r}_3-\underline{r}_1|^3}(x_3-x_1)\\ \frac{Gm}{|\underline{r}_2-\underline{r}_1|^3}(y_2-y_1)+\frac{Gm}{|\underline{r}_3-\underline{r}_1|^3}(y_3-y_1)\\ \frac{Gm}{|\underline{r}_1-\underline{r}_2|^3}(x_1-x_2)+\frac{Gm}{|\underline{r}_3-\underline{r}_2|^3}(x_3-x_2)\\ \frac{Gm}{|\underline{r}_1-\underline{r}_2|^3}(y_1-y_2)+\frac{Gm}{|\underline{r}_3-\underline{r}_2|^3}(y_3-y_2)\\ \frac{Gm}{|\underline{r}_1-\underline{r}_3|^3}(x_1-x_3)+\frac{Gm}{|\underline{r}_2-\underline{r}_3|^3}(x_2-x_3)\\ \frac{Gm}{|\underline{r}_1-\underline{r}_3|^3}(y_1-y_3)+\frac{Gm}{|\underline{r}_2-\underline{r}_3|^3}(y_2-y_3)\end{bmatrix}$$

<u>Initial Condition Calculation:</u>



Initial positions for each mass are shown in the diagram at the left.

The magnitude of the velocity vectors is

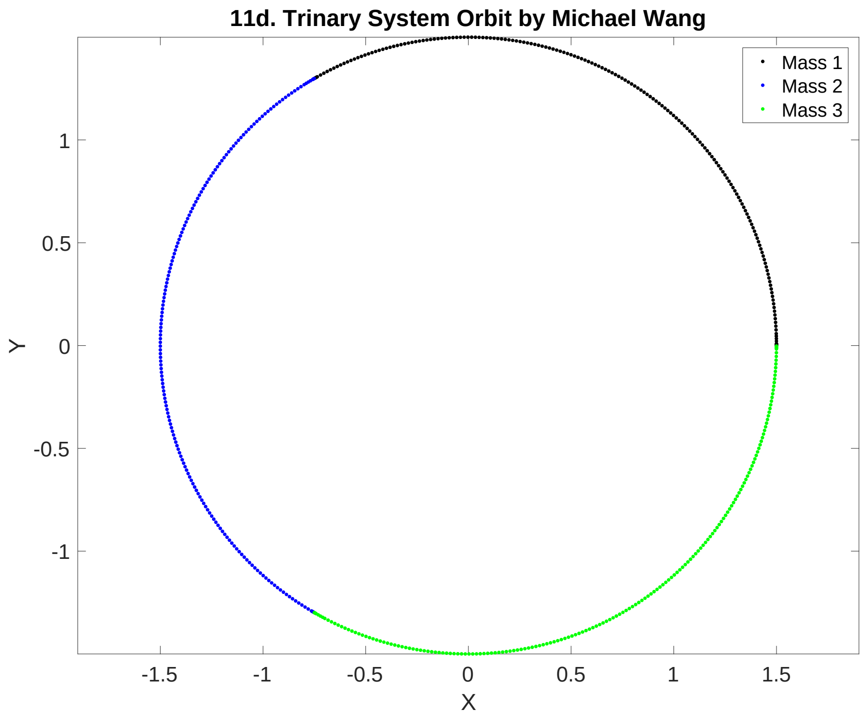
$$|v| = \dot{\theta} r = \sqrt{\frac{8Gm}{\sqrt{3}\,d^3}}\,\frac{d}{2}$$

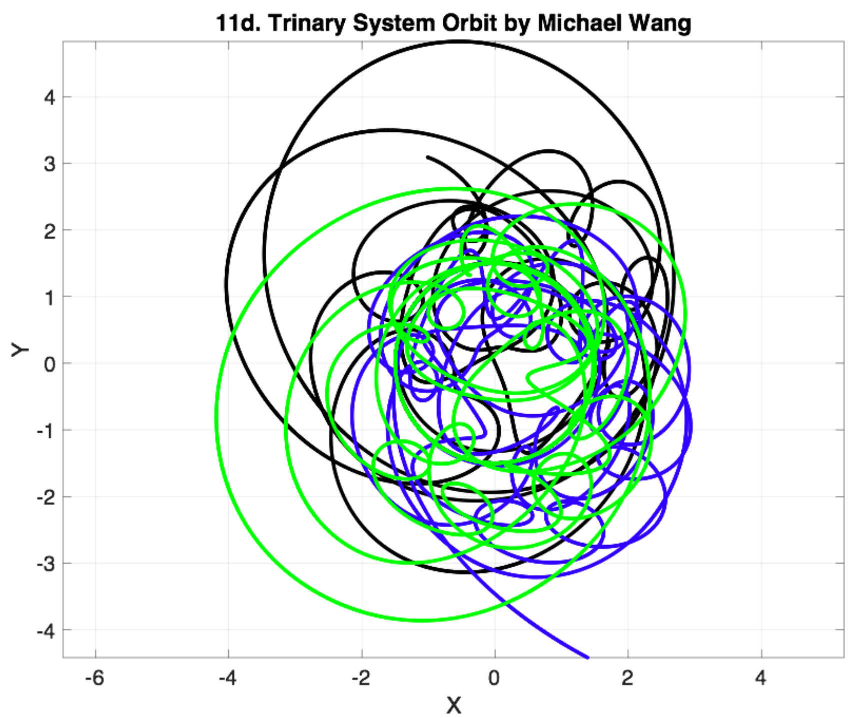
$$|v_1| = |v_2| = |v_3| = |v| \qquad \underline{v}_1 = |v|\,\underline{e}_2$$

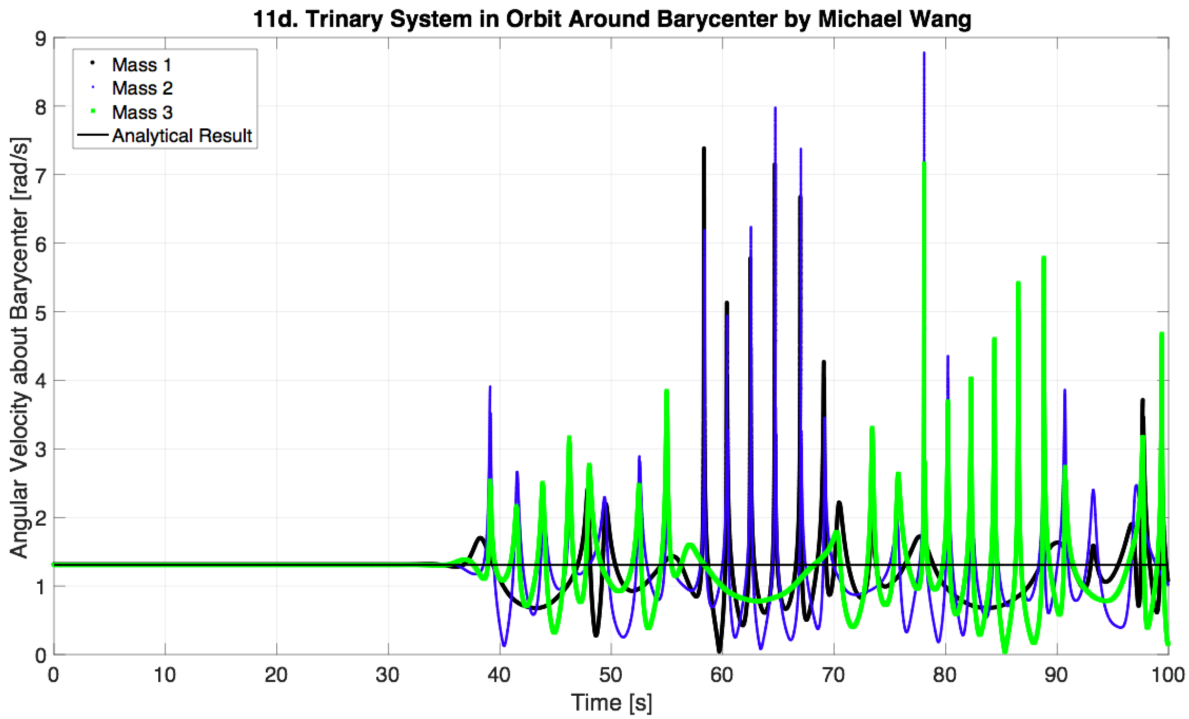
The x and y components of $\underline{v}_2$ and $\underline{v}_3$ can be calculated by rotating $\underline{v}_1$ counterclockwise by 120° (for $\underline{v}_2$) and by 240° (for $\underline{v}_3$).

> As shown in MATLAB plot, the simulated angular velocities match the value found in part (c)

Analytical	Simulated
1.308 rad/s	1.308 rad/s


11d. Trinary System Orbit by Michael Wang
Mass 1
Mass 2
Mass 3
1
0.5
0
-0.5
-1
Y
-1.5
-1
-0.5
0
0.5
1
1.5
X


11d. Trinary System Orbit by Michael Wang
4
3
2
1
0
-1
-2
-3
-4
Y
-6
-4
-2
0
2
4
X



System becomes unstable after 35 s.

```matlab
%% Problem 11b by Michael Wang
% parameters
p.G = 2000;
p.m1 = 800;
p.m2 = 400;
p.d = 300;

% Initial Conditions
% z = [r1;r2;theta1;theta2;r1dot;r2dot;theta1dot;theta2dot]
r10 = -(p.m2/(p.m1+p.m2))*p.d;
r1dot0 = 0;
theta10 = 0;
theta1dot0 = sqrt(p.G*(p.m1+p.m2)/(p.d^3));
r20 = (p.m1/(p.m1+p.m2))*p.d;
r2dot0 = 0;
theta20 = 0;
theta2dot0 = sqrt(p.G*(p.m1+p.m2)/(p.d^3));
z0 = [r10;r20;theta10;theta20;r1dot0;r2dot0;theta1dot0;theta2dot0];

% integration
tspan = [0,1000];
opts.RelTol = 1e-9;
opts.AbsTol = 1e-9;
[t,zarray] = ode45(@binary_system,tspan,z0,opts,p);
r1 = zarray(:,1); theta1 = zarray(:,3); theta1dot = zarray(:,7);
x1 = r1.*cos(theta1); y1 = r1.*sin(theta1);
r2 = zarray(:,2); theta2 = zarray(:,4); theta2dot = zarray(:,8);
x2 = r2.*cos(theta2); y2 = r2.*sin(theta2);

% plot
figure; hold on;
plot(t,theta1dot,'b*');
plot(t,theta2dot,'go');
plot([0,t(end)],[theta1dot0,theta1dot0],'k-');
grid on; box on;
title('11b. Binary System Angular Velocities by Michael Wang');
ylabel('Angular Velocity about Barycenter [rad/s]');
xlabel('Time [s]');
legend('Mass 1','Mass 2','Analytical Result','Location','Best');

figure; hold on;
plot(x1,y1,'k.');
plot(x2,y2,'b.');
grid on; axis equal; box on;
title('11b. Binary System Orbit by Michael Wang');
ylabel('Y');
xlabel('X');
legend('Mass 1','Mass 2');

%% Problem 11d. by Michael Wang
% parameters
p.m = 1;
p.G = 10;
p.d = 3;

% Initial Conditions
% z = [x1; y1; x2; y2; x3; y3; x1dot; y1dot; x2dot; y2dot; x3dot; y3dot]
r0 = p.d/2;
rdot0 = 0;
theta0 = 0;
% thetadot0 = sqrt(16*p.G*p.m*0.5236/(3*p.d^3));
thetadot0 = sqrt(8*p.G*p.m/(sqrt(3)*p.d^3));
```

```matlab
z0 = [r0; 0; -r0*cos(deg2rad(60)); r0*sin(deg2rad(60)); -r0*cos(deg2rad(60)); -
r0*sin(deg2rad(60));...
    0; r0*thetadot0; -r0*thetadot0*sin(deg2rad(60)); -r0*thetadot0*cos(deg2rad(60));...
    r0*thetadot0*sin(deg2rad(60)); -r0*thetadot0*cos(deg2rad(60))];

% integration
% tspan = [0,1.6];
tspan = [0,100];
opts.RelTol = 1e-9;
opts.AbsTol = 1e-9;
% [t,zarray] = leapfrog(@trinary_system,tspan,z0,100000,p);
[t,zarray] = ode45(@trinary_system,tspan,z0,opts,p);
x1 = zarray(:,1);
y1 = zarray(:,2);
x2 = zarray(:,3);
y2 = zarray(:,4);
x3 = zarray(:,5);
y3 = zarray(:,6);
x1dot = zarray(:,7);
y1dot = zarray(:,8);
x2dot = zarray(:,9);
y2dot = zarray(:,10);
x3dot = zarray(:,11);
y3dot = zarray(:,12);
theta1dot = zeros(length(t),1); theta2dot = zeros(length(t),1); theta3dot =
zeros(length(t),1);
for i = 1:length(t)
    theta1dot(i) = norm([x1dot(i), y1dot(i)])/r0;
    theta2dot(i) = norm([x2dot(i), y2dot(i)])/r0;
    theta3dot(i) = norm([x3dot(i), y3dot(i)])/r0;
end

% plot
figure; hold on;
plot(t,theta1dot,'ko','MarkerSize',2);
plot(t,theta2dot,'b*','MarkerSize',2);
plot(t,theta3dot,'gs','MarkerSize',2);
plot([0,t(end)],[thetadot0,thetadot0],'k-');
grid on; box on;
title('11d. Trinary System in Orbit Around Barycenter by Michael Wang');
ylabel('Angular Velocity about Barycenter [rad/s]');
xlabel('Time [s]');
legend('Mass 1','Mass 2','Mass 3','Analytical Result','Location','Best');

figure; hold on;
plot(x1,y1,'k.','MarkerSize',10);
plot(x2,y2,'b.','MarkerSize',10);
plot(x3,y3,'g.','MarkerSize',10);
grid on; axis equal; box on;
title('11d. Trinary System Orbit by Michael Wang');
ylabel('Y');
xlabel('X');
```

```
function zdot = binary_system(t,z,p)
% z = [r1;r2;theta1;theta2;r1dot;r2dot;theta1dot;theta2dot]
% unpack parameters
G = p.G;
m1 = p.m1;
m2 = p.m2;
d = p.d;
r1 = z(1);
r2 = z(2);
theta1 = z(3);
theta2 = z(4);
r1dot = z(5);
r2dot = z(6);
theta1dot = z(7);
theta2dot = z(8);


% dynamics
d = abs(r1) + abs(r2);
r1dotdot = G*m2/(d^2) + r1*theta1dot^2;
theta1dotdot = -2*r1dot*theta1dot/r1;
r2dotdot = -G*m1/(d^2) + r2*theta2dot^2;
theta2dotdot = -2*r2dot*theta2dot/r2;

% pack parameters
zdot = [r1dot;r2dot;theta1dot;theta2dot;r1dotdot;r2dotdot;theta1dotdot;theta2dotdot];
end
```

```
function zdot = trinary_system(t,z,p)
% z = [x1; y1; x2; y2; x3; y3; x1dot; y1dot; x2dot; y2dot; x3dot; y3dot]
% unpack parameters
G = p.G;
m = p.m;
d = p.d;
r = z(1);
x1 = z(1);
y1 = z(2);
x2 = z(3);
y2 = z(4);
x3 = z(5);
y3 = z(6);
x1dot = z(7);
y1dot = z(8);
x2dot = z(9);
y2dot = z(10);
x3dot = z(11);
y3dot = z(12);

% dynamics
% rdotdot = -8*G*m*0.5236/(3*d^2) + r*thetadot^2;
% thetadotdot = -2*rdot*thetadot/r;
d21 = norm([x2, y2] - [x1, y1]);
d31 = norm([x3, y3] - [x1, y1]);
d32 = norm([x3, y3] - [x2, y2]);
x1dotdot = G*m*(x2 - x1)/(d21^3) + G*m*(x3 - x1)/(d31^3);
y1dotdot = G*m*(y2 - y1)/(d21^3) + G*m*(y3 - y1)/(d31^3);
x2dotdot = G*m*(x1 - x2)/(d21^3) + G*m*(x3 - x2)/(d32^3);
y2dotdot = G*m*(y1 - y2)/(d21^3) + G*m*(y3 - y2)/(d32^3);
x3dotdot = G*m*(x1 - x3)/(d31^3) + G*m*(x2 - x3)/(d32^3);
y3dotdot = G*m*(y1 - y3)/(d31^3) + G*m*(y2 - y3)/(d32^3);

% pack parameters
zdot = [x1dot; y1dot; x2dot; y2dot; x3dot; y3dot; x1dotdot; y1dotdot; x2dotdot; y2dotdot;
x3dotdot; y3dotdot];
end
```